

# *Principles of Oral Surgery*

## *OMFS 711*

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# *Learning Objectives & Competency Statements*

## **Learning objectives**

- **Perform a thorough pre-operative assessment including appropriate consultations, medical risk assessment, premedication, and incorporating various modifications to treatment. ( competencies 2,3,4,5,6,8)**
- **Identify indications and contraindications and procedures for surgical extraction of erupted, partially, and fully impacted third molars. ( competencies 2,3,4,5,6,8)**
- **Provide sound post-surgical management and effectively evaluate and treat post-operative complications ( competencies 2,3,4,5,6,8)**

# *Principles of Surgery and wound Repair*

Human tissues have genomically predetermined characteristics that define normal responses to injury. Because the response to injury is predictable, principles of surgery have evolved to help optimize the wound-healing environment guided by basic and clinical research. We will be presenting the evidence-based principles of surgical practice that have been found to be successful not only for oral surgery, but also for surgery in all sites of the body.

The important decisions concerning a surgical procedure should be made well before the administration of anesthesia commences. The decision to perform surgery should be the result of careful patient examination. The surgeon first identifies the various signs and symptoms and relevant historical information; then using clinical experience the surgeon establishes the relationship between the individual problems for which surgical intervention may be indicated.

## *Basic Necessities for Surgery*

Little difference exists between the basic necessities required for oral surgery and those required for the proper performance of other aspects of dentistry. The two principal requirements are

- (1) Adequate visibility
- (2) Assistance.

Although visibility may seem too obvious to mention as a requirement for performing surgery, clinicians often underestimate its importance, especially when the unexpected occurs. Adequate visibility depends on the following three factors: (1) adequate access, (2) adequate light, and (3) a surgical field free of excess blood and other fluids and debris.

# *Aseptic Technique*

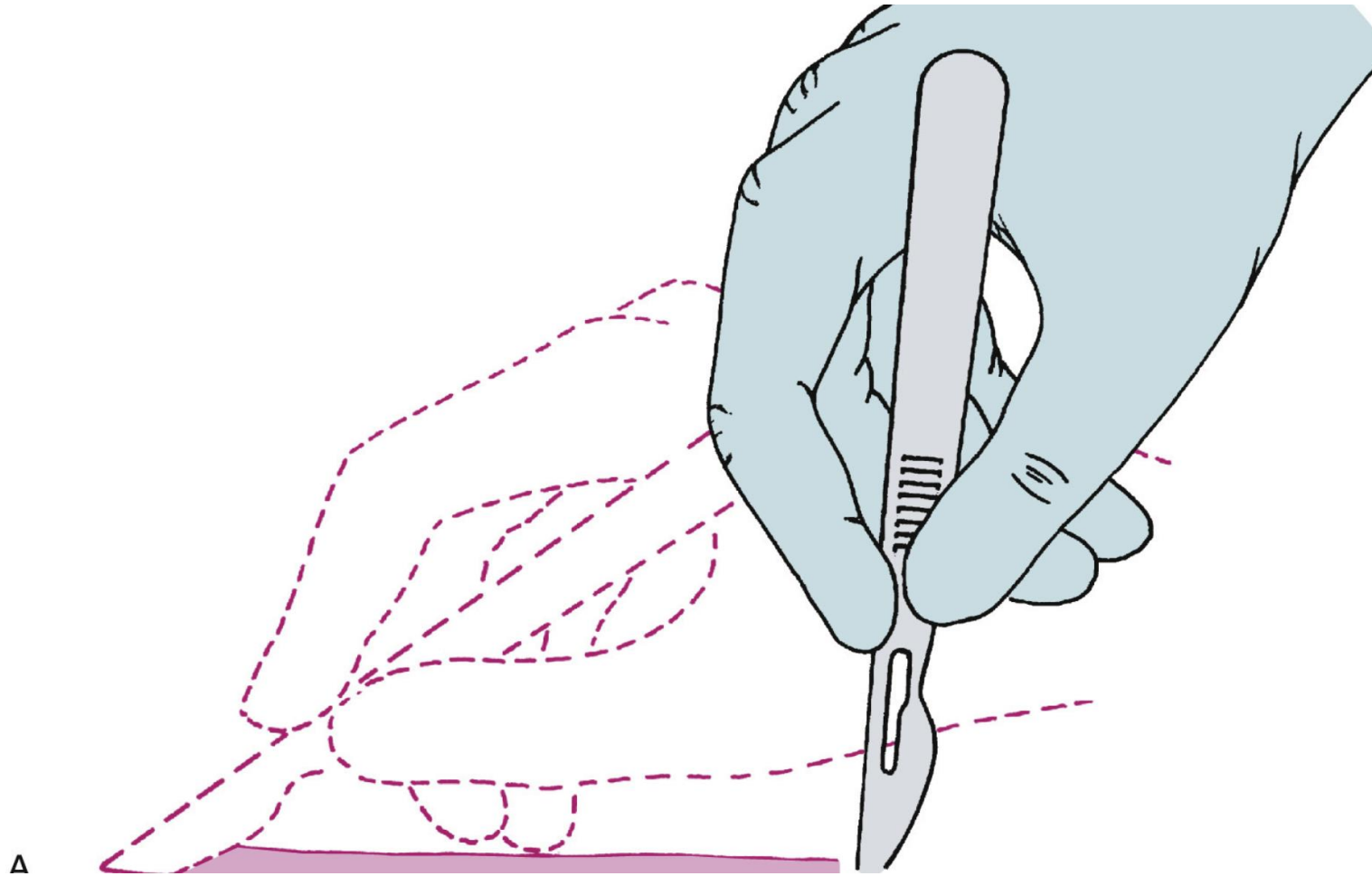
Aseptic technique is used to minimize wound contamination by pathogenic microbes. This important surgical principle is discussed in detail in following presentations.

# *Incisions*

Many oral and maxillofacial surgical procedures necessitate incisions.

The first principle:

is that a sharp blade of the proper size and shape should be used. A sharp blade allows the surgeon to make incisions precisely, without causing unnecessary injury caused by repeated strokes. The rate at which a blade dulls depends on the resistance of tissues through which the blade cuts. Bone and ligamentous tissues dull blades much more rapidly than does buccal mucosa. Therefore the surgeon should change the blade whenever the scalpel does not seem to be incising easily.



The second principle:

Is that a firm, continuous stroke should be used when incising. Repeated, tentative strokes increase the amount of damaged tissue within a wound and the amount of bleeding; these impair wound healing and visibility. Long, continuous strokes are preferable to short, interrupted ones

## The third principle:

Is that the surgeon should carefully avoid accidentally cutting important structures when incising. Each patient's microanatomy is unique. Therefore to avoid unintentionally cutting large vessels or nerves, the surgeon must incise only deeply enough to define the next major layer when making incisions close to where major vessels, ducts, and nerves run. Vessels can be more easily controlled before they are completely divided, and important nerves can usually be freed from adjacent tissue and retracted away from the area to be incised. In addition, when using a scalpel, the surgeon must remain focused on the location of the blade to avoid inadvertently cutting structures such as the lips when moving the scalpel into and out of the mouth.

## The fourth principle:

Is that incisions through epithelial surfaces that the surgeon plans to reapproximate should be made with the blade held perpendicular to the epithelial surface. This angle produces squared wound edges that are easier to reorient properly during suturing and are less susceptible to necrosis of the wound edges as a result of wound edge ischemia.

## The fifth principle:

Is that incisions in the oral cavity should be properly placed. Incisions through attached gingiva and over healthy bone are more desirable than those through unattached gingiva and over unhealthy or missing bone. Properly placed incisions allow the wound margins to be sutured over intact, healthy bone that is at least a few millimeters away from the damaged bone, thereby providing support for the healing wound. Similarly, when possible, incisions over prominences such as the canine eminence are best to avoid because the pressure on the closed wound from the prominence may interfere with wound healing. Incisions placed near the teeth for extractions should be made in the gingival sulcus, unless the clinician thinks that it is necessary to excise the marginal gingiva or to leave the marginal gingiva untouched.

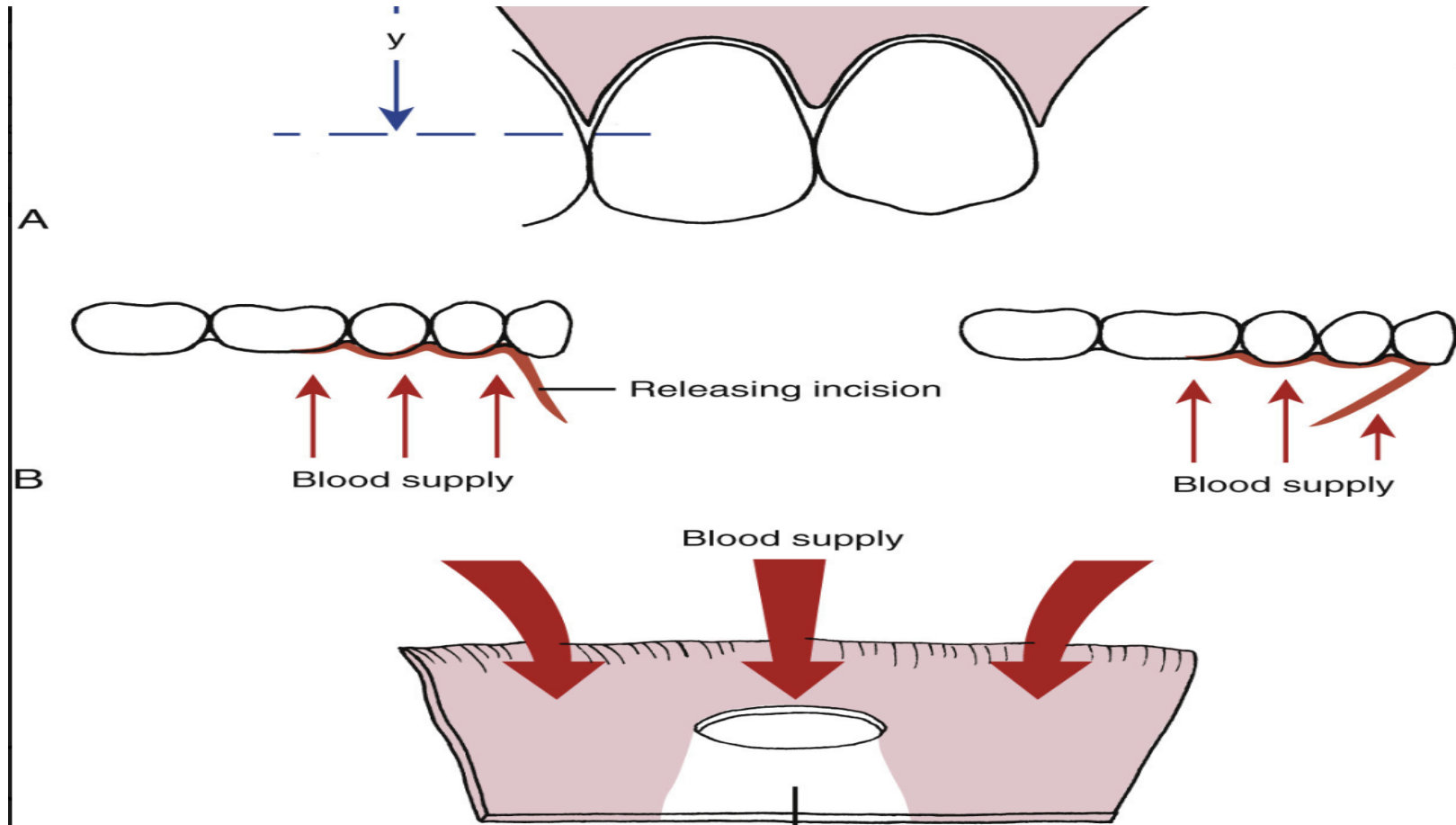
# *Flap Design*

Surgical flaps are made to gain surgical access to an area or to move tissue from one place to another. Several basic principles of flap design must be followed to prevent the primary complications of flap surgery: necrosis, dehiscence, and tearing.

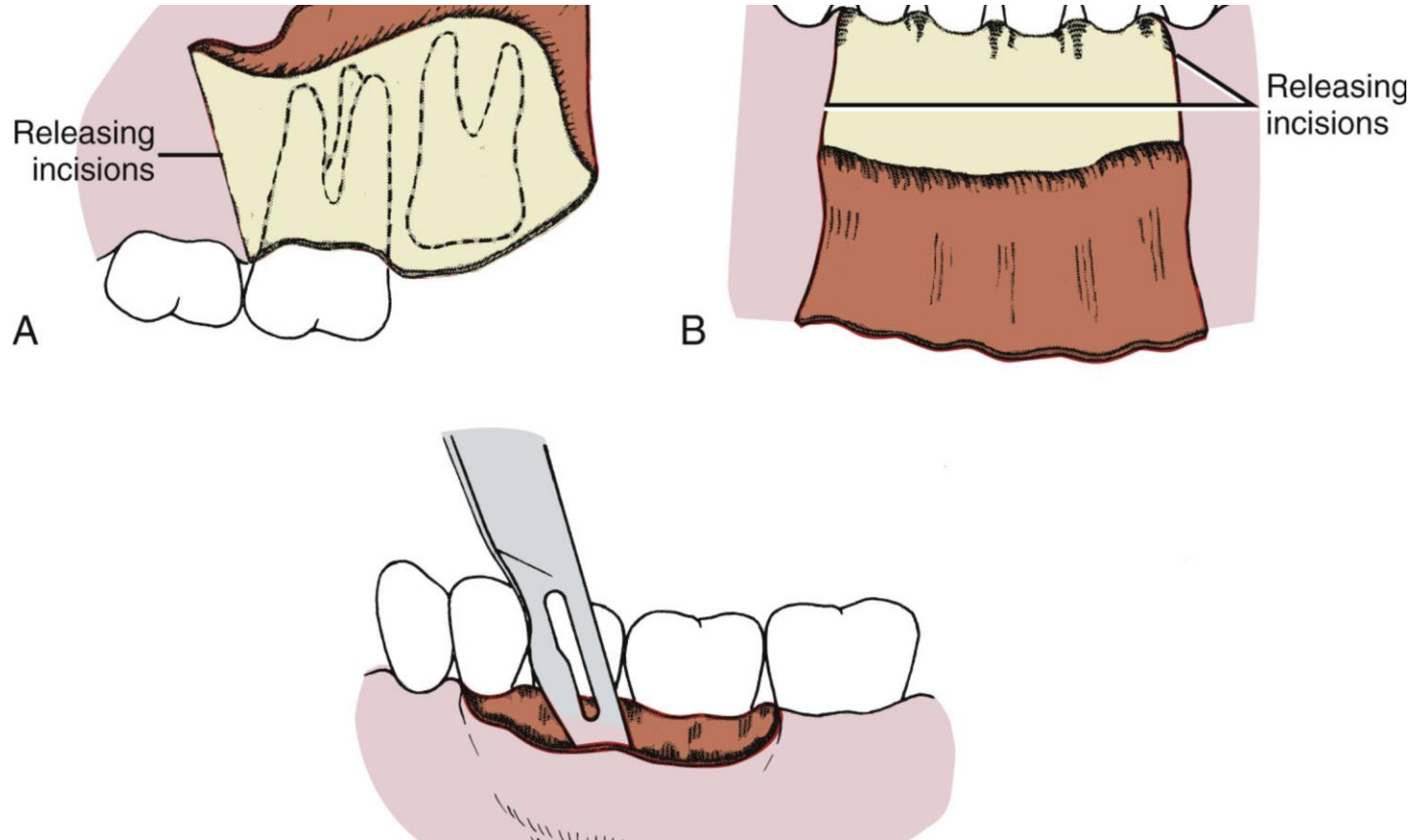
## *Prevention of Flap Necrosis*

Flap necrosis can be prevented if the surgeon attends to four basic flap design principles:

- (1) The height of a flap should never be greater than the base, unless a major artery is present in the base. Preferably, flaps should have sides that run parallel to each other or converge moving from the base to the apex of the flap.
- (2) (2) Generally, the height of a flap should be no more than twice the width of the base. The width of the base should preferably be greater than the height of the flap



- (3) When possible, an axial blood supply should be included in the base of the flap. For example, a flap in the palate should be based toward the greater palatine artery, when possible.
- (4) The base of flaps should not be excessively twisted, stretched, or grasped with anything that might damage vessels because these maneuvers could compromise the blood supply feeding and draining the flap, as well as delicate lymphatics.



## *Tissue Handling*

The difference between an acceptable and an excellent surgical outcome often rests on how the surgeon handles the tissues. The use of proper incision and flap design techniques plays a role; however, tissue also must be handled carefully. Excessive pulling or crushing, extremes of temperature, desiccation, or the use of unphysiologic chemicals easily damages tissue.

The surgeon should use care whenever touching tissue. When tissue forceps are used, they should not be pinched together too tightly; rather, they should be used to delicately hold tissue. When possible, toothed forceps or tissue hooks should be used to hold tissue



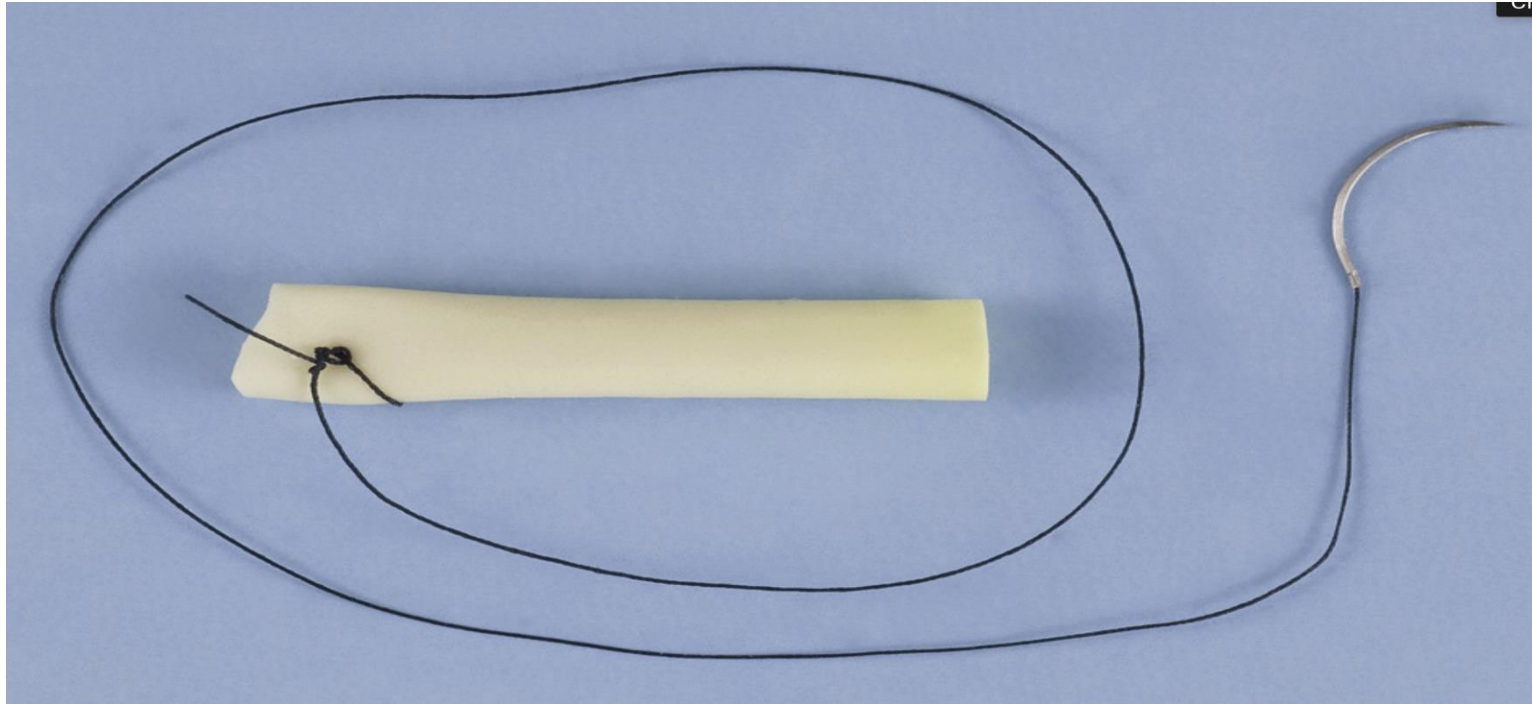
## *Hemostasis*

Prevention of excessive blood loss during surgery is important for preserving a patient's oxygen-carrying capacity. However, maintaining meticulous hemostasis during surgery is necessary for other important reasons. One is the decreased visibility that uncontrolled bleeding creates. Even high-volume suctioning cannot keep a surgical field completely dry, particularly in the well-vascularized oral and maxillofacial regions. Another problem bleeding causes is the formation of hematomas (collections of blood under soft tissue). Hematomas place pressure on wounds, decreasing vascularity; they increase tension on the wound edges; and they act as culture media, potentiating the development of wound infections.

## *Dead Space Management*

Dead space in a wound is any area that remains devoid of tissue after closure of the wound. Dead space is created by removing tissue in the depths of a wound and by not reapproximating all tissue planes during closure. Dead space in a wound usually fills with blood, which creates a hematoma with a high potential for infection.

# *Sutured drain*



## *Decontamination and Debridement*

Bacteria invariably contaminate all wounds that are open to the external or oral environment. Because the risk of infection rises with the increased size of an inoculum, one way to lessen the chance of wound infection is to decrease the bacterial count. This is easily accomplished by repeatedly irrigating the wound during surgery and closure. Irrigation, particularly when delivered under pressure, dislodges bacteria and other foreign materials and rinses them out of the wound.

## *Patient General Health and Wound Healing*

Proper wound healing depends on a patient's ability to resist infection, to provide essential nutrients for use as building materials, and to carry out reparative cellular processes. Numerous medical conditions impair a patient's ability to resist infection and heal wounds. These include conditions that establish a catabolic state of metabolism, that impede oxygen or nutrient delivery to tissues, or that require administration of drugs or physical agents that interfere with immunologic or wound-healing cells. Examples of diseases that induce a catabolic metabolic state include poorly controlled type 1 diabetes mellitus, end-stage renal or hepatic disease, and malignant diseases.

# *Wound Repair*

An important aspect of any surgical procedure is the preparation of the wound for proper healing. A thorough understanding of the biology of normal tissue repair is therefore valuable for individuals intending to perform surgery.

Tissue injury can be caused by pathologic conditions or by traumatic events. The dental surgeon has some control over pathologic tissue damage such as the likelihood of a wound infection. In addition, the surgeon can favorably or unfavorably alter the amount and severity of traumatically induced tissue injury and thus contribute to promoting or impeding wound healing.

**Causes of Tissue Damage,**

**Wound Repair,**

Epithelialization,  
Stages of Wound Healing,  
*Inflammatory Stage,*  
*Fibroplastic Stage,*  
*Remodeling Stage,*

**Surgical Significance of Wound-Healing Concepts,**

Factors That Impair Wound Healing,  
*Foreign Material,*  
*Necrotic Tissue,*  
*Ischemia,*  
*Tension,*  
Healing by Primary, Secondary, and Tertiary Intention,  
Healing of Extraction Sockets,  
Bone Healing,  
Implant Osseointegration,  
Facial Neuropathology of Traumatic Origin,  
*Nerve Healing,*  
*Classification,*

#### Box 4.1

### Causes of Tissue Damage

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#### Physical

- Compromised blood flow
- Crushing
- Desiccation
- Incision
- Irradiation
- Overcooling
- Overheating

#### Chemical

- Agents with unphysiologic pH
- Agents with unphysiologic tonicity
- Proteases
- Vasoconstrictors
- Thrombogenic agents

## *Epithelialization*

Injured epithelium has a genetically programmed regenerative ability that allows it to reestablish its integrity through proliferation, migration, and a process known as contact inhibition. In general, any edge of normal epithelium will begin and continue to migrate (by proliferation of germinal epithelial cells that advance the free edge forward) until it comes into contact with another free edge of epithelium, where it is signaled to stop growing laterally. Note that the other epithelium can be a different type of epithelium.

An example of the sometimes detrimental effect of the process of contact inhibition controlling epithelialization occurs when an opening is accidentally made into a maxillary sinus during tooth extraction.

If the epithelium of both the sinus wall and the oral mucosa is injured, it begins to proliferate in both areas. In this case, the first free epithelial edge the sinus epithelium may contact is oral mucosa, thereby creating an oroantral fistula

## *Stages of Wound Healing*

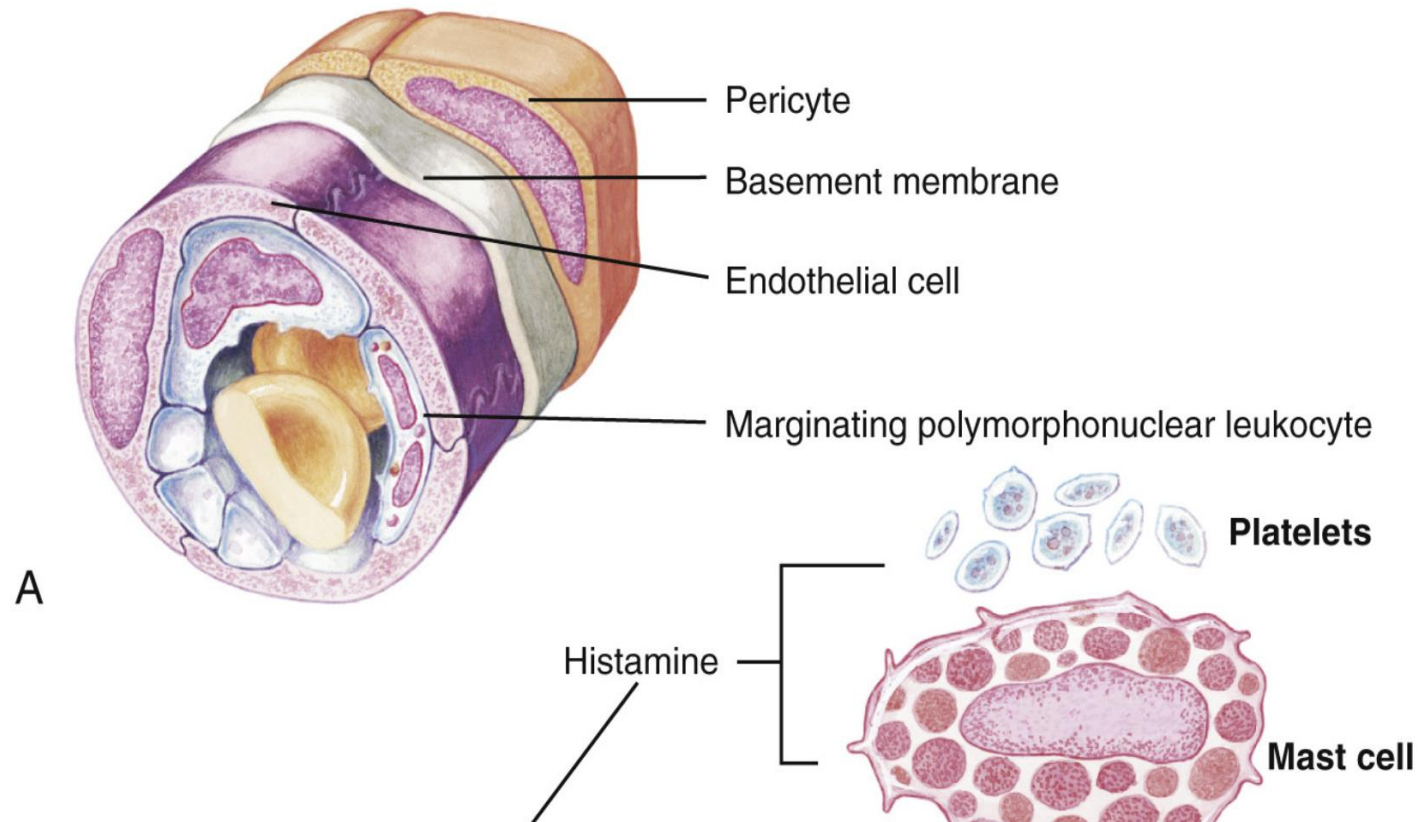
Regardless of the cause of nonepithelial tissue injury, a stereotypical process is initiated and, if able to proceed unimpeded, works to restore tissue integrity. This process is called wound healing. The process has been divided into basic stages that, although not mutually exclusive, take place in this sequence. These three basic stages are

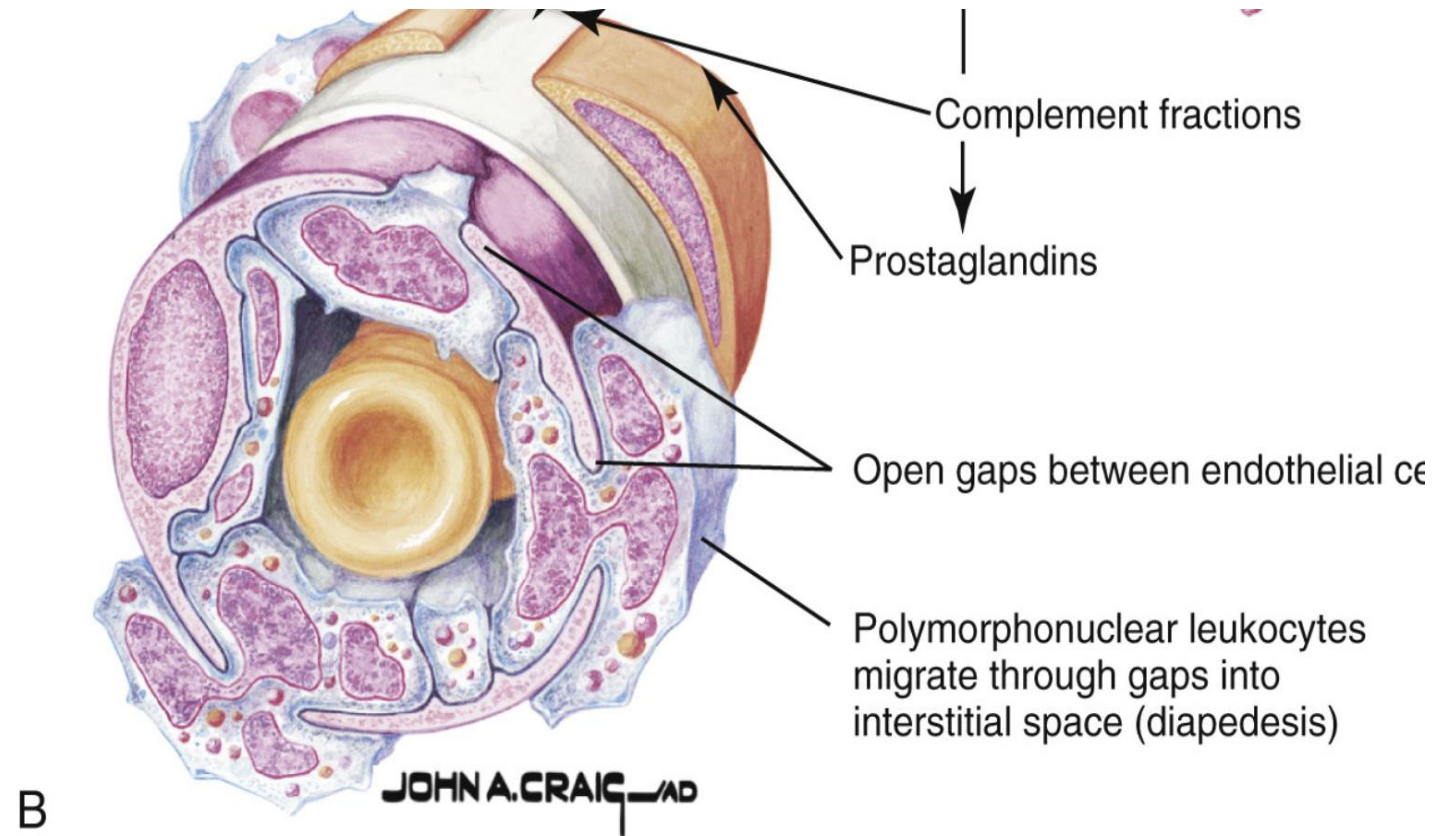
- (1) Inflammatory stage
- (2) Fibroplastic stage
- (3) Remodeling stage

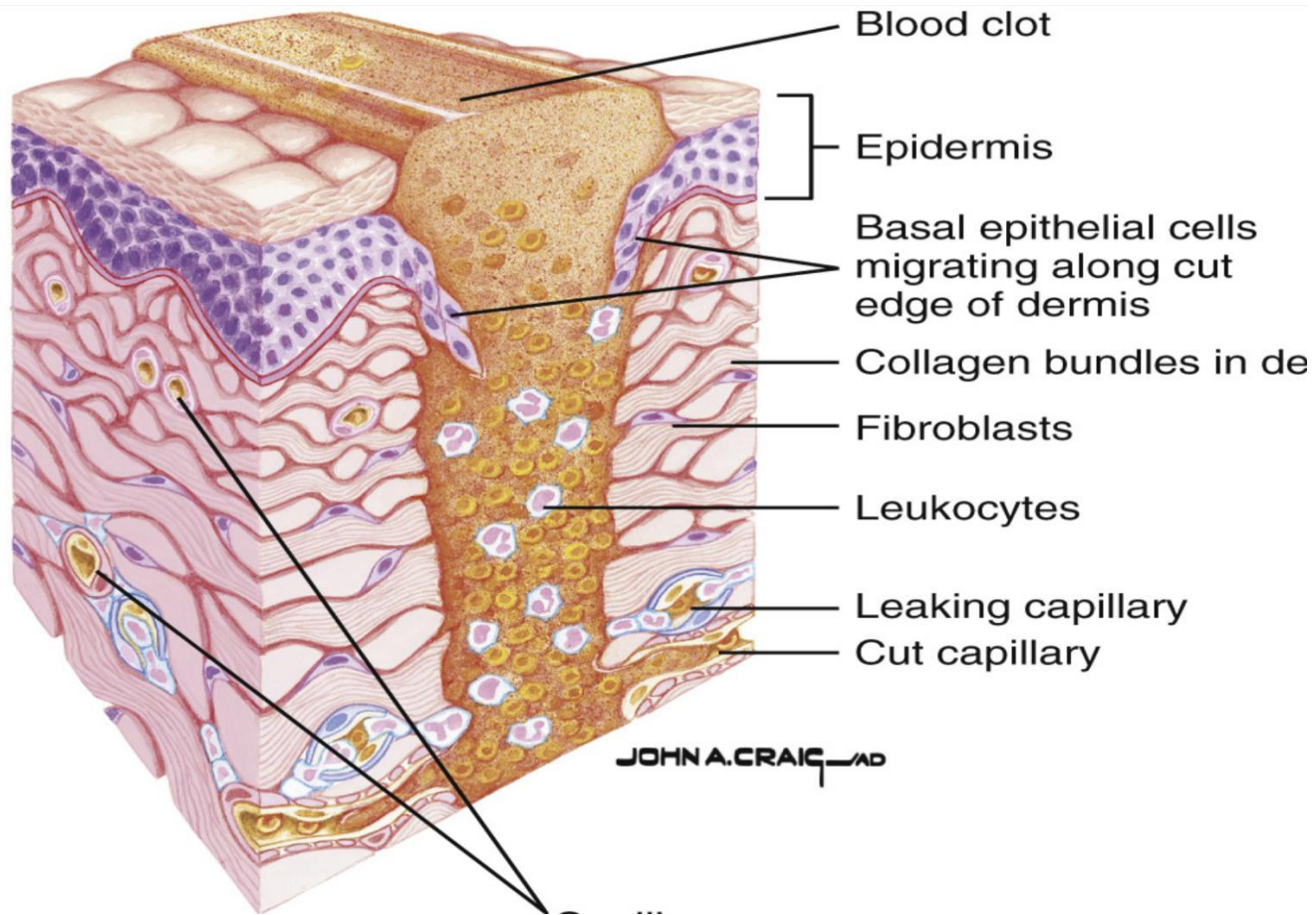
## *Inflammatory Stage*

The inflammatory stage begins the moment tissue injury occurs and, in the absence of factors that prolong inflammation, lasts 3 to 5 days. The inflammatory stage has two phases: (1) vascular and (2) cellular. The vascular events set in motion during inflammation begin with an initial vasoconstriction of disrupted vessels as a result of normal vascular tone. The vasoconstriction slows blood flow into the area of injury, promoting blood coagulation.

The vasoconstriction slows blood flow into the area of injury, promoting blood coagulation. Within minutes, histamine and prostaglandins E1 and E2, elaborated by white blood cells, cause vasodilation and open small spaces between endothelial cells, which allows plasma to leak and leukocytes to migrate into interstitial tissues. Fibrin from the transudated plasma causes lymphatic obstruction, and the transudated plasma—aided by obstructed lymphatic vessels—accumulates in the area of injury, functioning to dilute contaminants. This fluid collection is called edema .







## *Fibroplastic Stage*

The strands of fibrin, which are derived from blood coagulation, crisscross wounds forming a latticework on which fibroblasts begin laying down ground substance and tropocollagen. This is the fibroplastic stage of wound repair. The ground substance consists of several mucopolysaccharides, which act to cement collagen fibers together. The fibroblasts transform local and circulating pluripotential mesenchymal cells that begin tropocollagen production on the third or fourth day after tissue injury.

The fibrin network is also used by new capillaries, which bud from existing vessels along the margins of the wound and run along fibrin strands to cross the wound., the wound at the end of the fibroplastic stage will be stiff because of an excessive amount of collagen, erythematous because of the high degree of vascularization, and able to withstand 70% to 80% as much tension as uninjured tissue. Clinically, the wound at the end of the fibroplastic stage will be stiff because of an excessive amount of collagen, erythematous because of the high degree of vascularization, and able to withstand 70% to 80% as much tension as uninjured tissue

## *Factors That Impair Wound Healing*

Four factors can impair wound healing in an otherwise healthy individual:

- (1) foreign material.
- (2) necrotic tissue.
- (3) ischemia.
- (4) wound tension.

## *Healing by Primary, Secondary, and Tertiary Intention*

Clinicians use the terms primary intention and secondary intention to describe the two basic methods of wound healing. In healing by primary intention, the edges of a wound in which there is no tissue loss are placed and stabilized in essentially the same anatomic position they held before injury and are allowed to heal. Strictly speaking, healing by primary intention is only a theoretical ideal. However, the term is generally used to designate wounds in which the edges are closely reapproximated. Therefore healing occurs more rapidly, with a lower risk of infection, and with less scar formation than in wounds allowed to heal by secondary intention. Examples of wounds that heal by primary intention include well-repaired lacerations or incisions and well-reduced bone fractures.

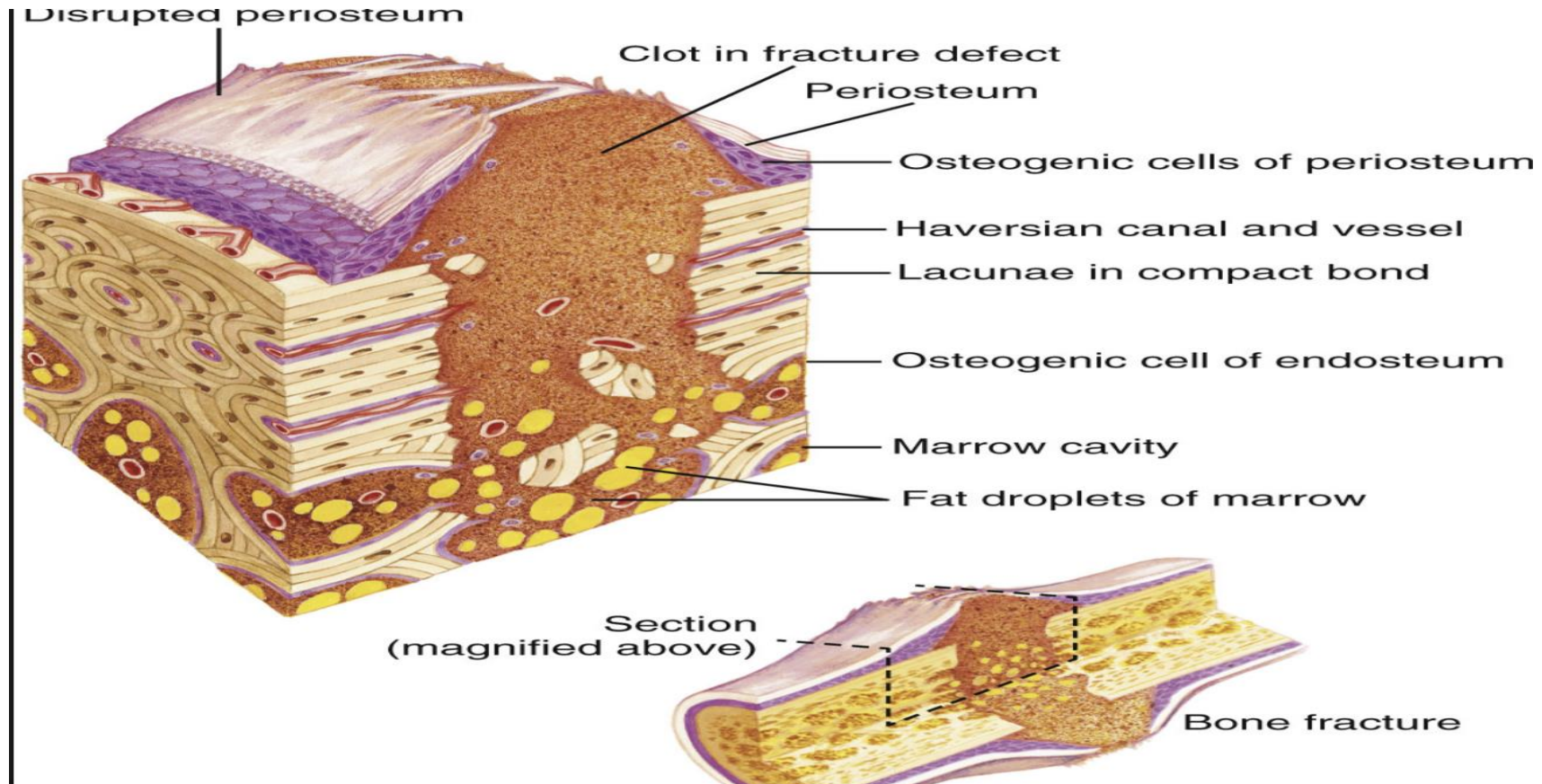
In contrast, healing by secondary intention implies that a gap is left between the edges of an incision or laceration or between bone or nerve ends after repair, or it implies that tissue loss has occurred in a wound that prevents approximation of wound edges. These situations require a large amount of epithelial migration, collagen deposition, contraction, and remodeling during healing. Healing is slower and produces more scar tissue than is the case with healing by primary intention. Examples of wounds that heal by secondary intention include extraction sockets, poorly reduced fractures, deep ulcers, and large avulsive injuries of any soft tissue.

Some surgeons use the term tertiary intention to refer to the healing of wounds through the use of tissue grafts to cover large wounds and bridge the gap between wound edges.

# *Bone Healing*

The events that occur during normal wound healing of soft tissue injuries (e.g., inflammation, fibroplasia, and remodeling) also take place during the repair of an injured bone. However, in contrast to soft tissues, osteoblasts and osteoclasts are also involved to reconstitute and remodel the damaged ossified tissue. Osteogenic cells (osteoblasts) important to bone healing are derived from the following three sources: (1) periosteum, (2) endosteum, and (3) circulating pluripotential mesenchymal cells. Under normal conditions, the fibrous tissue, including the callus, ossifies.

During the remodeling stage, bone that was haphazardly produced is resorbed by osteoclasts, and osteoblasts lay down new bone directed to resist low-grade tensions placed on the bone



Two factors are important to proper bone healing: (1) vascularity and (2) immobility. The fibrous connective tissue that forms in a bony fracture site requires a high degree of vascularity (which carries blood with a normal oxygen content) for eventual ossification. If vascularity or oxygen supplies are sufficiently compromised, cartilage, instead of bone, forms. Furthermore, if vascularity or oxygen supplies are poor, the fibrous tissue does not chondrify or ossify.

Placing bone under continuous or repeated cycles of some tension stimulates continued osteoblastic bone formation. Bone is formed perpendicular to lines of tension to help withstand the forces placed on it. This is the basis of the functional matrix concept of bone remodeling. However, excessive tension or torque placed on a healing fracture site produces mobility at the site.

# *Implant Osseointegration*

The discovery of osseointegration in the 1960s forced a reexamination of traditional concepts of wound healing. Before acceptance of these findings, it was thought that the body would eventually expel any foreign material placed through an epithelial surface. Expulsion would occur as the epithelium bordering the foreign material migrated down along the interface with the foreign material, finally fully surrounding the portion of the foreign body protruding into the body and causing the material to be completely external to the epithelial barrier. For a dental implant, this meant eventual loosening and loss of the implant.

The reasons why epithelium does not continue to migrate when it meets a bone and implant interface are still unclear. Nonetheless, dentistry has used this aberration in normal wound-healing principles to provide integrated metal posts (implants) that are useful to stabilize dental prostheses. Wound healing around dental implants involves the two basic factors: (1) healing of bone to the implant and (2) healing of alveolar soft tissue to the implant. Dental implants made of pure titanium are used in the discussion of healing around dental implants; similar healing occurs around properly placed implants made of other inert materials. Keeping forces off the implant prevents movement along the healing bone and implant interface during the critical portion of the healing period.

Finally, the surface to which bone is intended to attach must be reasonably free of surface contaminants. Such contaminants including bacteria, oil, glove powder, foreign metals, and foreign proteins should be minimized. The surface of an implant intended to osseointegrate should not be handled with bare or gloved fingers or forceps made of a metal different from the implant and must not have retained machine oil or detergent.

## *Facial Neuropathology of Traumatic Origin*

Injuries to sensory nerves of the maxillofacial region occasionally occur as the result of facial fractures, during the treatment of impacted teeth or oral pathologic conditions, or when maxillofacial reconstructive surgery is performed. Fortunately, most injured nerves spontaneously recover. However, in the past, little was done to treat persistent sensory nerve disorders. Advances in the understanding of how nerves heal and in the surgical means of repairing peripheral nerves provide patients with the possibility of partially or fully regaining normal nerve function.

The two branches of the trigeminal nerve injured most commonly, for which the altered sensation is clinically significant, are (1) the inferior alveolar-mental nerve and (2) the lingual nerve. When the inferior alveolar-mental nerve is injured, the usual causes are the following:

1. Mandibular (body) and angle fractures
2. Preprosthetic surgical procedures including implant placement
3. Sagittal split osteotomy surgery
4. Mandibular resection for oral neoplasms
5. Removal of impacted lower third molars
6. Local anesthetic injection

Research and clinical experience have shown that surgical intervention to repair damaged nerves is more successful when performed soon after the injury has occurred, thus an understanding of the various types of nerve damage, especially their prognoses, is important because it enables the clinician to decide when referral for peripheral nerve surgery is warranted.

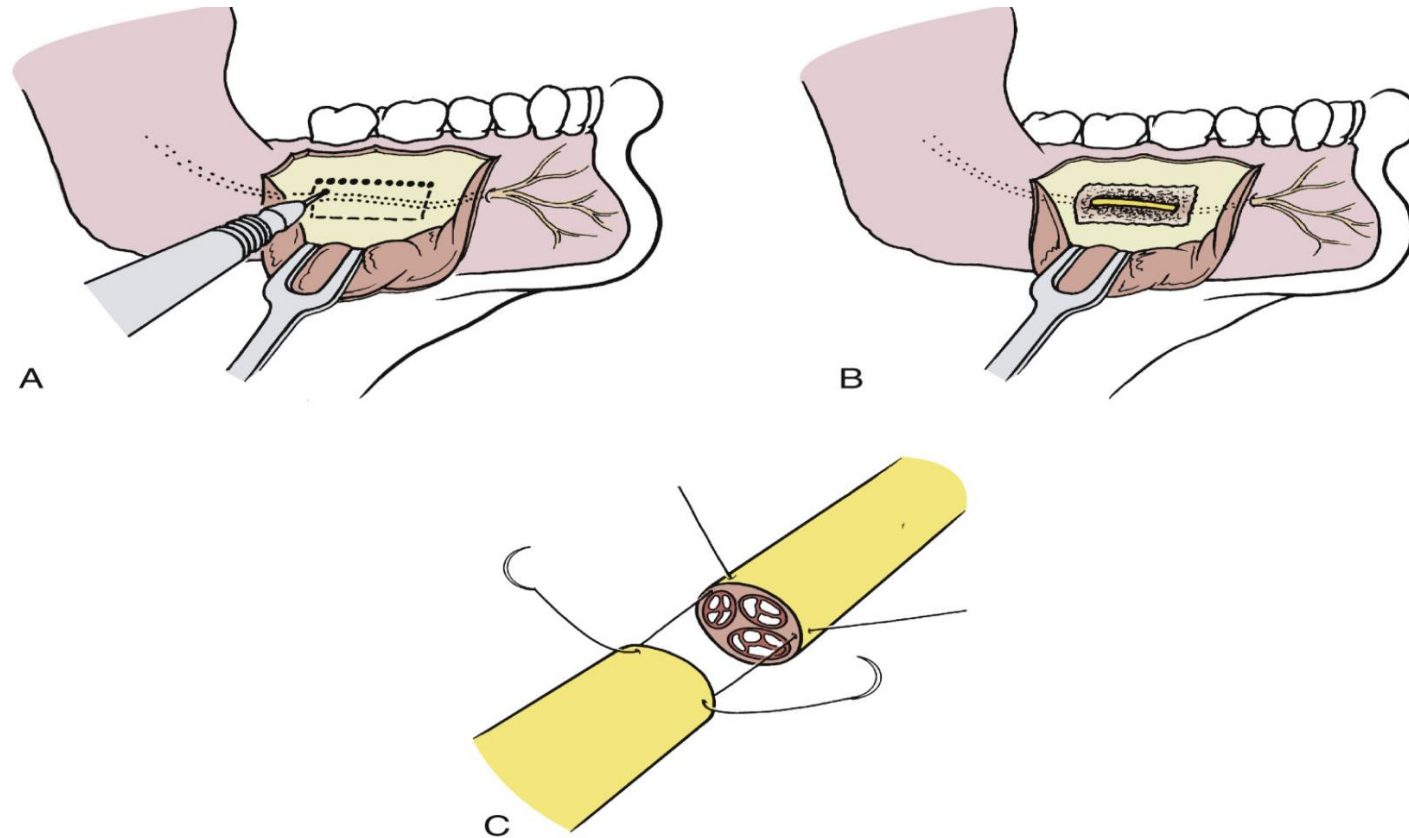
The three types of nerve injuries are

- (1) neurapraxia,
- (2) Axonotmesis
- (3) neurotmesis

# *Nerve Repair*

When there is a lack of spontaneous neurosensory regeneration due to neuroma formation, microneurosurgery may be required to achieve functional sensory recovery. or an inferior alveolar nerve injury, access must be obtained via a bony osteotomy. This approach may provide a decompression of the nerve and allow inspection for a neuroma that may require resection. If a neuroma is identified, microneurosurgery is performed to resect the neuroma. The nerve ends are then repaired with very fine epineurial sutures. Finally, in some cases in patients with dysesthesia, systemic medications (e.g.,  $\gamma$ -aminobutyric acid agonists) may also be required to control unpleasant neuropathic symptoms.

# *Nerve Repair*



*Thank you*